

## SAFETY DATA SHEET

Creation Date 12-April-2013

Revision Date 04-May-2023

Revision Number 6

### 1. Identification

**Product Name** 2-Cyanoethyl-N,N,N',N'-tetraisopropylphosphorodiamidite

**Cat No. :** AC379740000; AC379740010; AC379740050; AC379740250

**CAS-No** 102691-36-1

**Synonyms** Bis(diisopropylamino)-2-cyanoethoxyphosphine

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

##### **Manufacturer**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

##### **Emergency Telephone Number**

For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                  |            |
|--------------------------------------------------|------------|
| <b>Flammable liquids</b>                         | Category 3 |
| <b>Acute oral toxicity</b>                       | Category 4 |
| <b>Skin Sensitization</b>                        | Category 1 |
| <b>Physical Hazards Not Otherwise Classified</b> | Category 1 |
| Risk of explosion if heated under confinement    |            |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Flammable liquid and vapor

Harmful if swallowed

May cause an allergic skin reaction  
Risk of explosion if heated under confinement



### Precautionary Statements

#### Prevention

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Keep cool. Protect from sunlight  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use non-sparking tools  
Take action to prevent static discharges  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/protective clothing/eye protection/face protection

#### Response

Explosion risk in case of fire  
IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
Rinse mouth  
If skin irritation or rash occurs: Get medical advice/attention  
Wash contaminated clothing before reuse  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

#### Storage

Store in a well-ventilated place. Keep cool

#### Disposal

Dispose of contents/container to an approved waste disposal plant

### Other Hazards

Harmful to aquatic life with long lasting effects

## 3. Composition/Information on Ingredients

| Component                                                             | CAS-No      | Weight % |
|-----------------------------------------------------------------------|-------------|----------|
| Phosphorodiamidous acid, tetrakis(1-methylethyl)-, 2-cyanoethyl ester | 102691-36-1 | <=100    |

## 4. First-aid measures

### General Advice

If symptoms persist, call a physician.

### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

|                                 |                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                                                                                                                                              |
| Ingestion                       | Clean mouth with water and drink afterwards plenty of water.                                                                                                                                                                                                                                                                              |
| Most important symptoms/effects | None reasonably foreseeable. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| Notes to Physician              | Treat symptomatically                                                                                                                                                                                                                                                                                                                     |

## 5. Fire-fighting measures

|                                  |                                                                                                                   |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media     | Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. Water mist may be used to cool closed containers. |
| Unsuitable Extinguishing Media   | No information available                                                                                          |
| Flash Point                      | 60 °C / 140 °F                                                                                                    |
| Method -                         | No information available                                                                                          |
| Autoignition Temperature         | No information available                                                                                          |
| Explosion Limits                 |                                                                                                                   |
| Upper                            | No data available                                                                                                 |
| Lower                            | No data available                                                                                                 |
| Sensitivity to Mechanical Impact | No information available                                                                                          |
| Sensitivity to Static Discharge  | No information available                                                                                          |

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Combustible material.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Oxides of phosphorus.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

Health  
2

Flammability  
2

Instability  
3

Physical hazards  
N/A

## 6. Accidental release measures

|                                      |                                                                                                                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Personal Precautions                 | Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.            |
| Environmental Precautions            | Should not be released into the environment. Do not flush into surface water or sanitary sewer system.                                                                        |
| Methods for Containment and Clean Up | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|          |                                                                                                                                                                                                                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Handling | Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges. |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Storage.**

Keep refrigerated. Keep under nitrogen. Keep away from heat, sparks and flame. Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong bases. Strong oxidizing agents.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

**Physical State**

Liquid

**Appearance**

Clear

**Odor**

No information available

**Odor Threshold**

No information available

**pH**

No information available

|                                        |                                       |
|----------------------------------------|---------------------------------------|
| Melting Point/Range                    | No data available                     |
| Boiling Point/Range                    | 98 - 100 °C / 208.4 - 212 °F @ 5 mmHg |
| Flash Point                            | 60 °C / 140 °F                        |
| Evaporation Rate                       | No information available              |
| Flammability (solid,gas)               | Not applicable                        |
| Flammability or explosive limits       |                                       |
| Upper                                  | No data available                     |
| Lower                                  | No data available                     |
| Vapor Pressure                         | No information available              |
| Vapor Density                          | No information available              |
| Specific Gravity                       | 0.949                                 |
| Solubility                             | No information available              |
| Partition coefficient; n-octanol/water | No data available                     |
| Autoignition Temperature               | No information available              |
| Decomposition Temperature              | No information available              |
| Viscosity                              | No information available              |
| Molecular Formula                      | C15 H32 N3 O P                        |
| Molecular Weight                       | 301.41                                |

## 10. Stability and reactivity

|                                  |                                                                                                                   |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                               |
| Stability                        | Moisture sensitive. heat sensitive.                                                                               |
| Conditions to Avoid              | Exposure to moisture. Heat, flames and sparks. Keep away from open flames, hot surfaces and sources of ignition.  |
| Incompatible Materials           | Strong bases, Strong oxidizing agents                                                                             |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Nitrogen oxides (NO <sub>x</sub> ), Oxides of phosphorus |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                          |
| Hazardous Reactions              | None under normal processing.                                                                                     |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | May cause sensitization by skin contact                                                  |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                                                             | CAS-No      | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------------------------------------------------------------|-------------|------------|------------|------------|------------|------------|
| Phosphorodiamidous acid, tetrakis(1-methylethyl)-, 2-cyanoethyl ester | 102691-36-1 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                   |                          |
|-------------------|--------------------------|
| Mutagenic Effects | No information available |
|-------------------|--------------------------|

|                                                   |                                                                                                                                                                                                                                                                               |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reproductive Effects</b>                       | No information available.                                                                                                                                                                                                                                                     |
| <b>Developmental Effects</b>                      | No information available.                                                                                                                                                                                                                                                     |
| <b>Teratogenicity</b>                             | No information available.                                                                                                                                                                                                                                                     |
| <b>STOT - single exposure</b>                     | None known                                                                                                                                                                                                                                                                    |
| <b>STOT - repeated exposure</b>                   | None known                                                                                                                                                                                                                                                                    |
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                                                                                                                      |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing:<br>Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                                                                                                                                                                      |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                                                |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

|                                      |                           |
|--------------------------------------|---------------------------|
| <b>Persistence and Degradability</b> | No information available  |
| <b>Bioaccumulation/ Accumulation</b> | No information available. |
| <b>Mobility</b>                      | No information available. |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

|                             |                                                                       |
|-----------------------------|-----------------------------------------------------------------------|
| <b>DOT</b>                  | Not regulated                                                         |
| <b>Proper Shipping Name</b> | consumer commodity                                                    |
| <b>Technical Name</b>       | Phosphorodiamidous acid, tetrakis(1-methylethyl)-, 2-cyanoethyl ester |
| <b>TDG</b>                  | Not regulated                                                         |
| <b>IATA</b>                 |                                                                       |
| <b>UN-No</b>                | UN1993                                                                |
| <b>Proper Shipping Name</b> | Flammable liquid, n.o.s.                                              |
| <b>Hazard Class</b>         | 3                                                                     |
| <b>Packing Group</b>        | III                                                                   |
| <b>IMDG/IMO</b>             |                                                                       |
| <b>UN-No</b>                | UN1993                                                                |
| <b>Proper Shipping Name</b> | Flammable liquid, n.o.s.                                              |
| <b>Hazard Class</b>         | 3                                                                     |
| <b>Packing Group</b>        | III                                                                   |

## 15. Regulatory information

**All of the components in the product are on the following Inventory lists:** X = listed U.S.A. (TSCA) Canada (DSL/NDL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (KECL) China (IECSC) Japan (ENCS) Philippines (PICCS) Taiwan (TCSI) Japan (ISHL) New Zealand (NZIoC) Japan (ISHL)

### International Inventories

| Component                                                             | CAS-No      | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS | ELINCS | NLP |
|-----------------------------------------------------------------------|-------------|-----|------|------|-----------------------------------------------|--------|--------|-----|
| Phosphorodiamidous acid, tetrakis(1-methylethyl)-, 2-cyanoethyl ester | 102691-36-1 | -   | -    | -    | -                                             | -      | -      | -   |

| Component                                                             | CAS-No      | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------------------------------------------------------------------|-------------|-------|------|------|------|------|------|-------|-------|
| Phosphorodiamidous acid, tetrakis(1-methylethyl)-, 2-cyanoethyl ester | 102691-36-1 | -     | -    | -    | X    | X    | -    | X     | -     |

**Legend:**

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

Not applicable

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                                                             | CAS-No      | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------------------------------------------------------|-------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Phosphorodiamidous acid, tetrakis(1-methylethyl)-, 2-cyanoethyl ester | 102691-36-1 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                                             | CAS-No      | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Phosphorodiamidous acid, tetrakis(1-methylethyl)-, 2-cyanoethyl ester | 102691-36-1 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information****Prepared By**

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**Creation Date**

12-April-2013

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04-May-2023

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**Revision Summary**

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**